

Homework Unit 4 (ET4260)

Semiconductor Sensors

Instructions:

Document your calculations properly with derivations and screenshots when needed.

Please submit your files in word format to facilitate feedback. Latex-generated PDF files are also accepted but not preferred.

State here your NAME and STUDENT NUMBER.

Part A: Hall Sensors

Task 1 a):

Considering the material properties given in table 1, which of these intrinsic semiconductors is most suitable to make high-sensitivity Hall sensors? Explain why.

Table 1: Material Properties Si, GaAs, Ge

Property	Si	GaAs	Ge
Atoms (cm ⁻³)	5.0×10^{22}	4.42×10^{22}	4.42×10^{22}
Atomic weight	28.09	144.63	72.60
Crystal structure	Diamond	Zincblende	Diamond
Density (g/cm ³)	2.33	5.32	5.33
Lattice constant (Å)	5.43	5.65	5.65
Melting point (°C)	1415	1238	937
Dielectric constant	11.7	13.1	16.0
Bandgap energy (eV)	1.12	1.42	0.66
Electron affinity, χ (V)	4.01	4.07	4.13
Effective density of states in conduction band, N_c (cm ⁻³)	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
Effective density of states in valence band, N_v (cm ⁻³)	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
Intrinsic carrier concentration (cm ⁻³)	1.5×10^{10}	1.8×10^6	2.4×10^{13}
Mobility (cm ² /V-s)			
Electron, μ_n	1350	8500	3900
Hole, μ_p	480	400	1900

Answer:

- the sensitivity of a Hall sensor is given by

$$S_I = \frac{1}{n \cdot q \cdot t} \cdot r \cdot G$$

$$S_V = \frac{\mu}{\left(\frac{L}{W}\right)_{\text{eff}}} \cdot r \cdot G$$

- effective L/W as well as the geometry factor are determined by geometry
- scattering factor is also dependent on material, but no information given here
- given the table data, a material with either low intrinsic carrier concentration and or high carrier mobility will give highly sensitive Hall sensors
- here: n-type GaAs has best properties

Task 1 b):

Which device parameter links the current-related sensitivity S_I and the voltage-related sensitivity S_V of a Hall sensor to each other? Why must this parameter be considered as well in designing a Hall sensor?

Answer:

They are related by the devices input resistance:

$$S_I = \frac{V_{\text{Hall}}}{I_{\text{bias}} \cdot B}$$
$$S_V = \frac{V_{\text{Hall}}}{V_{\text{bias}} \cdot B} = \frac{V_{\text{Hall}}}{I_{\text{bias}} \cdot R_{\text{in}} \cdot B} = \frac{S_I}{R_{\text{in}}}$$

The (input) resistance of a hall device is relevant since it is usually equal to the output resistance (symmetrical Hall plates) or at least related, which will determine the noise level of the sensor (and influence the SNR).

Task 1 c):

When comparing integrated (including circuitry, such as biasing and read-out) Hall devices, which sensitivity is a reasonable figure of merit?

Answer:

In an integrated circuit, biasing is limited by the system voltages. A Hall sensor with a high input resistance can have a very high current-related sensitivity, but high currents cannot be provided due to voltage limitations. In many cases ICs have fixed voltage levels available, so that the voltage related sensitivity is a better measure.

Task 1 d):

Consider the following GaN-Hall devices from paper [1] given in table 2:

Table 2: GaN Hall Sensor Properties

Property	Gen 1	Gen 2	Gen 3	Si
S_I (V/A/T)	68	90	34	300
S_V (mV/V/T)	76	57	34	60
Heterostructure Barrier	AlGaN	AlGaN	InAlN	N/A
Distance between contacts (μm)	200	340	340	100
Shape	Oct	CGC	CGC	Oct
R_{in}, R_{out} (Ω)	870	1560	1030	5350 (0V) to 6500 (2V)
R nonlinearity (%/V)	-10	0.5	0.5	5
Min Offset (<1V)(μT)	500	0.5	2	3
Max Offset (<3V)(μT)	3000	3	4	15
Thermal Noise Floor (nT/ $\sqrt{\text{Hz}}$)	50	89	120	200
Corner Frequency (1V) (kHz)	100 (projected)	0.6	1	7.0

For the Gen 2 devices, the sheet resistance was $R_{\square} = 430 \Omega/\square$.

- Calculate $\left(\frac{L}{W}\right)_{\text{eff}}$ and the geometry factor for the Gen 2 device.
- Given Gen 3 has the same geometry as Gen 2, calculate its' sheet resistance.
- Calculate the sheet carrier density ($n \cdot t$) for Gen 2 and Gen3 in #carriers/cm⁻² as well as the carrier mobility (cm²/Vs). Assume $r = 1$ and $e = 1.602 \times 10^{-19} \text{ C}$.
- Additional task: Compare the sheet resistance and carrier mobility of the silicon device to the values calculated above, assuming $\left(\frac{L}{W}\right)_{\text{eff}} = \sqrt{2}$.

Answer:

i.

$$R = R_{\square} \cdot \left(\frac{L}{W}\right)_{\text{eff}}$$

$$\left(\frac{L}{W}\right)_{\text{eff}} = \frac{R}{R_{\square}} = \frac{1560 \, \Omega}{430 \, \Omega/\square} = 3.63$$

$$G \approx \frac{\left(\frac{L}{W}\right)_{\text{eff}}^2}{\sqrt{\left(\frac{L}{W}\right)_{\text{eff}}^4 + \frac{1}{2}\left(\frac{L}{W}\right)_{\text{eff}}^2 + 4}} = \frac{3.63^2}{\sqrt{3.63^4 + \frac{1}{2} \cdot 3.63^2 + 4}} = 0.971$$

ii.

$$R_{\square} = \frac{R}{\left(\frac{L}{W}\right)_{\text{eff}}} = \frac{1030 \, \Omega}{3.63} = 283.75 \, \Omega/\square$$

iii.

sheet carrier density:

$$S_I = \frac{1}{n \cdot e \cdot t} \cdot r \cdot G$$

$$n \cdot t = \frac{1}{e \cdot S_I} \cdot r \cdot G$$

$$(n \cdot t)_{\text{Gen2}} = \frac{1}{1.602 \times 10^{-19} \, \text{C} \cdot 90 \, \text{V/A/T}} \cdot 1 \cdot 0.971 = 6.73 \times 10^{12} \, \text{cm}^{-2}$$

$$(n \cdot t)_{\text{Gen3}} = \frac{1}{1.602 \times 10^{-19} \, \text{C} \cdot 34 \, \text{V/A/T}} \cdot 1 \cdot 0.971 = 1.78 \times 10^{13} \, \text{cm}^{-2}$$

mobility:

$$S_V = \mu \cdot \frac{r \cdot G}{\left(\frac{L}{W}\right)_{\text{eff}}}$$

$$\mu = \frac{\left(\frac{L}{W}\right)_{\text{eff}} \cdot S_V}{r \cdot G}$$

$$\mu_{\text{Gen2}} = \frac{3.63 \cdot 0.057 \, \text{V/V/T}}{1 \cdot 0.971} = 2130.90 \, \frac{\text{cm}^2}{\text{Vs}}$$

$$\mu_{\text{Gen3}} = \frac{3.63 \cdot 0.034 \, \text{V/V/T}}{1 \cdot 0.971} = 1271.06 \, \frac{\text{cm}^2}{\text{Vs}}$$

iv.

$$R_{\square} = \frac{5350 \, \Omega}{\sqrt{2}} = 3783.01 \, \Omega/\square$$

$$G = \frac{\left(\frac{L}{W}\right)_{eff}^2}{\sqrt{\left(\frac{L}{W}\right)_{eff}^4 + \frac{1}{2}\left(\frac{L}{W}\right)_{eff}^2 + 4}} = \frac{2}{\sqrt{4 + 1 + 4}} = \frac{2}{3}$$

$$\mu = \frac{\sqrt{2} \cdot 0.06 \text{ V/V/T}}{1 \cdot 2/3} = 1272.79 \frac{\text{cm}^2}{\text{Vs}}$$

Part B: Photodetectors

Task 2 a):

- i. Calculate the silicon active device thickness to achieve 90 % and 99 % absorption of light at the wavelengths of 600 nm ($a = 4140 \text{ cm}^{-1}$) and 400 nm ($\alpha = 95200 \text{ cm}^{-1}$).
- ii. Calculate the relative change in quantum efficiency for the thicknesses you calculated for each wave length.
- iii. Calculate the quantum efficiency for each of the wavelengths for 99 % absorption. Assume $\mathcal{R}(400 \text{ nm}) = 0.485$, $\mathcal{R}(600 \text{ nm}) = 0.354$ and $\xi = 1$.

Answer:

i.

$$\phi(x) = \phi_0 e^{-\alpha \cdot x}$$

$$x = -\frac{1}{\alpha} \cdot \ln\left(\frac{\phi}{\phi_0}\right)$$

600 nm:

$$x_{0.90} = -\frac{1}{0.414 \text{ } \mu\text{m}^{-1}} \cdot \ln(0.1) = 5.56 \text{ } \mu\text{m}$$

$$x_{0.99} = -\frac{1}{0.414 \text{ } \mu\text{m}^{-1}} \cdot \ln(0.01) = 11.12 \text{ } \mu\text{m}$$

400 nm:

$$x_{0.90} = -\frac{1}{9.52 \text{ } \mu\text{m}^{-1}} \cdot \ln(0.1) = 0.24 \text{ } \mu\text{m}$$

$$x_{0.99} = -\frac{1}{9.52 \text{ } \mu\text{m}^{-1}} \cdot \ln(0.01) = 0.48 \text{ } \mu\text{m}$$

ii.

$$\eta = (1 - \mathcal{R}(\lambda)) \cdot \xi \cdot (1 - e^{-\alpha(\lambda) \cdot x})$$

$$\Delta\eta_{\text{rel}} = \frac{\eta_{0.99} - \eta_{0.90}}{\eta_{0.90}} \times 100 \% = \frac{(1 - e^{-\alpha(\lambda) \cdot x_{0.99}}) - (1 - e^{-\alpha(\lambda) \cdot x_{0.90}})}{(1 - e^{-\alpha(\lambda) \cdot x_{0.90}})} \times 100 \%$$

Since the statements $1 - e^{-\alpha(\lambda) \cdot x_{0.90}}$ and $1 - e^{-\alpha(\lambda) \cdot x_{0.99}}$ represent 90 % and 99 % absorption, one can write:

$$\Delta\eta_{\text{rel}} = \frac{0.99 - 0.90}{0.90} \times 100 \%$$

$$\Delta\eta_{\text{rel}} = \left(\frac{0.99}{0.90} - 1\right) \times 100 \% = 10 \%$$

iii.

$$\eta = (1 - \mathcal{R}(\lambda)) \cdot \xi \cdot (1 - e^{-\alpha(\lambda) \cdot x})$$

Since the statement $1 - e^{-\alpha(\lambda) \cdot x_{0.99}}$ represent 99 % absorption, one can write:

$$\eta = (1 - \mathcal{R}(\lambda)) \cdot \xi \cdot 0.99$$

600 nm:

$$\eta_{0.99} = (1 - 0.354) \cdot \xi \cdot 0.99 = 0.640$$

400 nm:

$$\eta_{0.99} = (1 - 0.485) \cdot \xi \cdot 0.99 = 0.510$$

Task 2 b):

Consider a piece of n-doped silicon as a photoconductor at room temperature with the following properties:

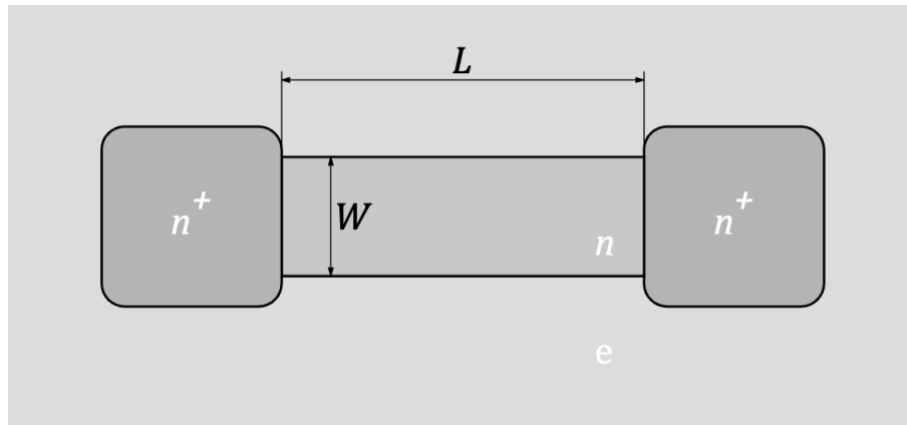


Figure 1: Integrated Photoconductor

Donor concentration of $N_D = 10^{17} \text{ cm}^{-3}$ (assume $n = N_D$) and mobilities $\mu_n = 700 \text{ cm}^2/\text{Vs}$ and $\mu_p = 240 \text{ cm}^2/\text{Vs}$, n-well depth of $t = 1 \text{ }\mu\text{m}$, resistance width of $W = 10 \text{ }\mu\text{m}$ and resistance length of $L = 30 \text{ }\mu\text{m}$.

The photoconductor is driven with a 1.8 V bias (only consider the conductor resistance).

- Calculate the current without optical excitation. The intrinsic carrier concentration is $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, the elementary charge is $e = 1.602 \times 10^{-19} \text{ C}$ and the Boltzmann constant is $k_B = 1.381 \times 10^{-23} \text{ J/K}$.
- Calculate the photocurrent for a photon flux of $\phi = \Phi/(L \cdot W) = 10^{16} \text{ s}^{-1}\text{m}^{-2}$. Assume $\eta = 0.515$ and $\tau = 100 \text{ ns}$.

iii. Calculate the conductor gain. How much does the gain change in the following situations:

a) the conductor is twice as long ($2L$)

b) the conductor is twice as wide ($2W$)

iv. How does the photocurrent change for the cases above?

Answer:

i.

$$I_0 = \sigma_0 \cdot E \cdot A = \sigma_0 \cdot \frac{V}{L} \cdot W \cdot t$$

The conductance of a semiconductor is given by:

$$\sigma_0 = e \cdot (\mu_n \cdot n + \mu_p \cdot p)$$

The concentration of holes and follows the equilibrium condition:

$$n \cdot p = n_i^2$$

Following $n = N_D$

$$p = \frac{n_i^2}{N_D} = \frac{(1.5 \times 10^{10} \text{ cm}^{-3})^2}{10^{17} \text{ cm}^{-3}} = 2.25 \times 10^3 \text{ cm}^{-3}$$

So that:

$$\sigma_0 = 1.602 \times 10^{-19} \text{ C} \cdot \left(700 \frac{\text{cm}^2}{\text{Vs}} \cdot 10^{17} \frac{1}{\text{cm}^3} + 240 \frac{\text{cm}^2}{\text{Vs}} \cdot 2.25 \times 10^3 \frac{1}{\text{cm}^3} \right)$$

$$\sigma_0 = 11.214 \frac{1}{\Omega \text{cm}} = 1121.4 \frac{1}{\Omega \text{m}}$$

That the initial current becomes:

$$I_0 = 1121.4 \frac{1}{\Omega \text{m}} \cdot \frac{1.8 \text{ V}}{30 \times 10^{-6} \text{ m}} \cdot 10 \times 10^{-6} \text{ m} \cdot 10^{-6} \text{ m}$$

$$I_0 = 672.84 \mu\text{A}$$

ii.

$$i_p = e \cdot \eta \cdot \phi \cdot \frac{\tau}{\tau_e} \cdot \left(1 + \frac{\mu_p}{\mu_n} \right)$$

$$i_p = e \cdot \eta \cdot \Phi \cdot L \cdot W \cdot \frac{\tau}{\tau_e} \cdot \left(1 + \frac{\mu_p}{\mu_n} \right)$$

The electron transit time is calculated by

$$\tau_e = \frac{L}{\mu_n \cdot E} = \frac{L}{\mu_n \cdot \frac{V}{L}} = \frac{(30 \times 10^{-6} \text{ m})^2}{700 \times 10^{-4} \frac{\text{m}}{\text{Vs}} \cdot 1.8 \text{ V}} = 7.14 \times 10^{-9} \text{ s}$$

And the photocurrent by

$$i_p = 1.602 \times 10^{-19} \text{ C} \cdot 0.515 \cdot \frac{10^{16}}{\text{s} \cdot \text{m}^2} \cdot 3 \cdot 10^{-5} \text{ m} \cdot 10^{-5} \text{ m} \cdot \frac{10^{-7} \text{ s}}{7.14 \times 10^{-9} \text{ s}} \left(1 + \frac{0.024 \frac{\text{m}^2}{\text{Vs}}}{0.07 \frac{\text{m}^2}{\text{Vs}}} \right)$$

$$i_p = 4.653 \times 10^{-12} \text{ A}$$

iii.

$$\Gamma_{ph} = \frac{\tau_p}{t_e} \left(1 + \frac{\mu_p}{\mu_n} \right)$$

The electron transit time is calculated by

$$\tau_e = \frac{L}{\mu_n \cdot E} = \frac{L}{\mu_n \cdot \frac{V}{L}} = \frac{(30 \times 10^{-6} \text{ m})^2}{700 \times 10^{-4} \frac{\text{m}}{\text{Vs}} \cdot 1.8 \text{ V}} = 7.14 \times 10^{-9} \text{ s}$$

The photoconductor gain is therefore given by

$$\Gamma_{ph} = \frac{10^{-7} \text{ s}}{7.14 \times 10^{-9} \text{ s}} \left(1 + \frac{240 \frac{\text{cm}^2}{\text{Vs}}}{700 \frac{\text{cm}^2}{\text{Vs}}} \right) = 18.81$$

iv.

for $2L$ conductor length the transit time becomes

$$t_n = \frac{(2 \cdot 30 \times 10^{-6} \text{ m})^2}{700 \times 10^{-4} \frac{\text{m}}{\text{Vs}} \cdot 1.8 \text{ V}} = 28.57 \times 10^{-9} \text{ s}$$

and the gain

$$\Gamma_{ph} = \frac{10^{-7} \text{ s}}{28.57 \times 10^{-9} \text{ s}} \left(1 + \frac{240 \frac{\text{cm}^2}{\text{Vs}}}{700 \frac{\text{cm}^2}{\text{Vs}}} \right) = 4.70$$

for $2W$ conductor width the gain remains the same.

v.

The photocurrent is given by:

$$i_p = e \cdot \eta \cdot \Phi \cdot L \cdot W \cdot \frac{\tau}{\frac{L^2}{\mu_n \cdot V}} \cdot \left(1 + \frac{\mu_p}{\mu_n} \right)$$

$$i_p = e \cdot \eta \cdot \Phi \cdot \frac{W}{L} \cdot \tau \cdot \mu_n \cdot V \cdot \left(1 + \frac{\mu_p}{\mu_n} \right)$$

So that for doubling the length, the photocurrent is halved and for doubling the width it is doubled.

Part C: Temperature Sensors

Task 3 a):

Consider the Wheatstone bridge depicted below. The resistances R_1 and R_4 are moderately n -doped poly-Si and have a negative TCR of $\alpha_{\text{neg}} = -3 \times 10^{-4} \text{ K}^{-1}$ and resistances R_2 and R_3 are highly n -doped poly-Si and thus have a positive TCR of $\alpha_{\text{pos}} = +5 \times 10^{-4} \text{ K}^{-1}$. All resistances are designed to have a value of $R = 1 \text{ k}\Omega$ at 300 K. Find the bias-related sensitivity of the bridge towards changes in temperature.

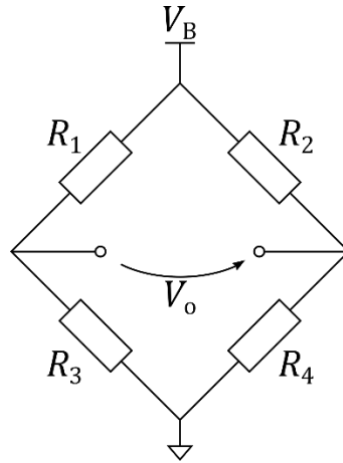


Figure 2: Wheatstone Bridge

Answer:

The output-Voltage can be calculated as difference between the center-potentials of the two voltage dividers:

$$V_1 = V_B \cdot \frac{R_3}{R_3 + R_1} = V_B \cdot \frac{R + \alpha_p \cdot R \cdot \Delta T}{R + \alpha_p \cdot R \cdot \Delta T + R + \alpha_n \cdot R \cdot \Delta T} = V_B \cdot \frac{1 + \alpha_p \cdot \Delta T}{2 + (\alpha_p + \alpha_n) \cdot \Delta T}$$

$$V_2 = V_B \cdot \frac{R_4}{R_4 + R_2} = V_B \cdot \frac{R + \alpha_n \cdot R \cdot \Delta T}{R + \alpha_n \cdot R \cdot \Delta T + R + \alpha_p \cdot R \cdot \Delta T} = V_B \cdot \frac{1 + \alpha_n \cdot \Delta T}{2 + (\alpha_p + \alpha_n) \cdot \Delta T}$$

$$V_o = V_1 - V_2 = V_B \frac{(\alpha_p - \alpha_n) \cdot \Delta T}{2 + (\alpha_p + \alpha_n) \cdot \Delta T} \approx \frac{V_B}{2} \cdot (\alpha_p - \alpha_n) \cdot \Delta T$$

For $2 \gg (\alpha_p + \alpha_n) \cdot \Delta T$ (which can be generally assumed if the temperature coefficients are either on the same order of magnitude or the temperature changes are moderate):

$$V_o \approx \frac{V_B}{2} \cdot (\alpha_p - \alpha_n) \cdot \Delta T$$

The sensitivity is given by:

$$S = \frac{1}{V_B} \cdot \frac{\delta V_o}{\delta T} \approx \frac{1}{2} \cdot (\alpha_p - \alpha_n) = \frac{1}{2} \cdot (5 + 3) \times 10^{-4} \text{ K}^{-1} = 4 \times 10^{-4} \text{ K}^{-1}$$

Exact Solution:

$$V_o = V_B \frac{(\alpha_p - \alpha_n) \cdot \Delta T}{2 + (\alpha_p + \alpha_n) \cdot \Delta T}$$

$$S = \frac{1}{V_B} \cdot \frac{\delta V_o}{\delta \Delta T} = \frac{d \cdot n' - d' \cdot n}{d^2}$$

$$S = \frac{(2 + (\alpha_p + \alpha_n) \cdot \Delta T) \cdot (\alpha_p - \alpha_n) - (\alpha_p + \alpha_n) \cdot (\alpha_p - \alpha_n) \cdot \Delta T}{(2 + (\alpha_p + \alpha_n) \cdot \Delta T)^2}$$

$$S = \frac{2 \cdot (\alpha_p - \alpha_n)}{(2 + (\alpha_p + \alpha_n) \cdot \Delta T)^2}$$

For the range of e.g. $-100 \text{ K} \leq \Delta T \leq 100 \text{ K}$ the sensitivity will be:

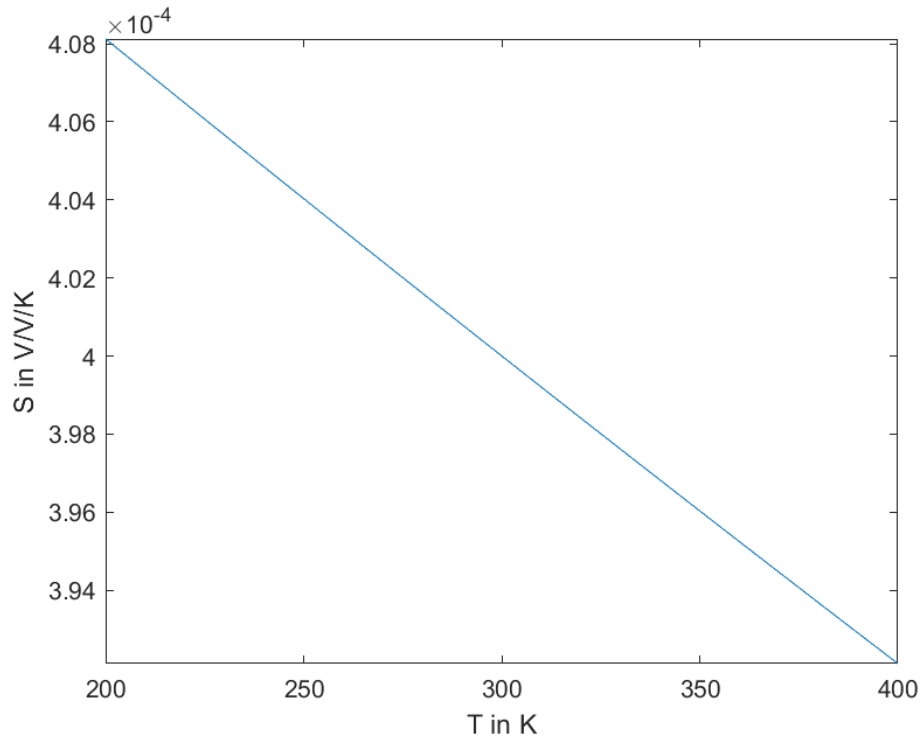


Figure 3: Wheatstone Bridge Sensitivity

Task 3 b):

Compare the statements for output voltage and output rms-noise voltage of a Wheatstone bridge ($R_1 = R_4 = R(1 + \alpha_{\text{neg}} \cdot \Delta T)$, $R_2 = R_3 = R(1 + \alpha_{\text{pos}} \cdot \Delta T)$). How will the SNR generally behave for changing temperature? Assume $\alpha_{\text{neg}} = -\alpha_{\text{pos}}$.

Additional task: plot the exact ($\alpha_{\text{neg}} \neq \alpha_{\text{pos}}$) SNR for $T_0 = 300 \text{ K} \leq T \leq 400 \text{ K}$, assume $ENBW = 1 \text{ kHz}$, $V_B = 5 \text{ V}$ and the R and α -values given in part a). $k_B = 1.381 \times 10^{-23} \text{ J/K}$.

Answer:

The SNR is given by

$$SNR = \frac{V_o}{v_{\text{rms}}}$$

or in dB as

$$SNR_{\text{dB}} = 20 \cdot \log\left(\frac{V_o}{v_{\text{rms}}}\right)$$

The output voltage for $\alpha_n = -\alpha_p$ is given by

$$V_o = V_B \cdot \alpha \cdot \Delta T$$

And the rms-noise by:

$$v_{\text{rms}} = \sqrt{4k_B T R_{\text{out}}} \cdot \sqrt{ENBW}$$

Where the output resistance is:

$$R_{\text{out}} = (R_1 || R_3) + (R_2 || R_4)$$

$$R_{\text{out}} = \frac{R_1 \cdot R_3}{R_1 + R_3} + \frac{R_2 \cdot R_4}{R_2 + R_4}$$

$$R_{\text{out}} = \frac{R(1 + \alpha_n \cdot \Delta T) \cdot R(1 + \alpha_p \cdot \Delta T)}{R(1 + \alpha_n \cdot \Delta T) + R(1 + \alpha_p \cdot \Delta T)} + \frac{R(1 + \alpha_p \cdot \Delta T) \cdot R(1 + \alpha_n \cdot \Delta T)}{R(1 + \alpha_p \cdot \Delta T) + R(1 + \alpha_n \cdot \Delta T)}$$

$$R_{\text{out}} = \frac{2R^2(1 + \alpha_n \cdot \Delta T) \cdot (1 + \alpha_p \cdot \Delta T)}{R((1 + \alpha_n \cdot \Delta T) + (1 + \alpha_p \cdot \Delta T))} = \frac{2R((1 + \alpha_n \cdot \Delta T) \cdot (1 + \alpha_p \cdot \Delta T))}{2 + (\alpha_n + \alpha_p)\Delta T}$$

Applying $\alpha_n = -\alpha_p$:

$$R_{\text{out}} = R \cdot (1 - \alpha_p^2 \Delta T^2)$$

Note: The extended calculation of rms-noise at the bridge output can be found in the Appendix.

It is fair to assume that $1 \gg \alpha_p^2 \cdot \Delta T^2$. For example in the temperature range of $-100 \text{ K} \leq \Delta T \leq 100 \text{ K}$ with $\alpha_p^2 = (5 \times 10^{-4} \text{ K}^{-1})^2 = 2.5 \times 10^{-7} \text{ K}^{-2}$ this term is within: $\alpha_p^2 \cdot \Delta T^2 \ll 2.5 \times 10^{-3}$. Therefore:

$$R_{\text{out}} \approx R$$

And

$$v_{\text{rms}} = \sqrt{4 \cdot k \cdot T \cdot R \cdot ENBW}$$

The output voltage and rms-noise show the following relation to temperature:

$$V_o \propto T$$

$$v_{\text{rms}} \propto \sqrt{T}$$

For an increasing temperature difference (actual temperature to reference temperature) the SNR will increase. Considering $SNR = V_o/v_{\text{rms}}$ the increase will be approximately be proportional to $\sqrt{T}$.

To plot the exact SNR, one needs to use the exact expressions for output voltage and rms-noise:

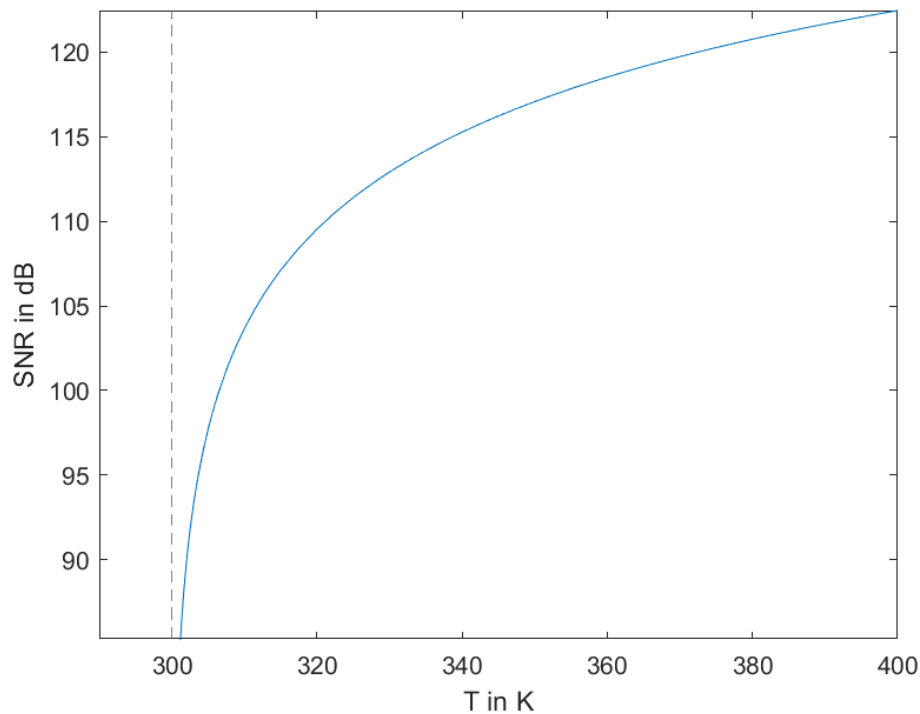
$$V_o = V_B \frac{(\alpha_p - \alpha_n) \cdot \Delta T}{2 + (\alpha_p + \alpha_n) \cdot \Delta T}$$

$$v_{\text{rms}} = \sqrt{8 \cdot k_B \cdot T \cdot R \frac{((1 + \alpha_n \cdot \Delta T) \cdot (1 + \alpha_p \cdot \Delta T))}{2 + (\alpha_n + \alpha_p) \Delta T} \cdot ENBW}$$

$$SNR_{\text{dB}} = 20 \cdot \log \left(\frac{\left(V_B \cdot \frac{(\alpha_p - \alpha_n) \cdot \Delta T}{2 + (\alpha_p + \alpha_n) \cdot \Delta T} \right)}{\sqrt{8 \cdot k_B \cdot R \frac{(1 + \alpha_n \cdot \Delta T) \cdot (1 + \alpha_p \cdot \Delta T)}{2 + (\alpha_n + \alpha_p) \Delta T} \cdot T \cdot ENBW}} \right)$$

$$SNR = 20 \cdot \log \left(\frac{\left(10 \text{ V} \cdot \frac{(5 \times 10^{-4} \text{ K}^{-1} + 3 \times 10^{-4} \text{ K}^{-1}) \cdot (T - T_0)}{2 + 2 \times 10^{-4} \text{ K}^{-1} \cdot (T - T_0)} \right)}{\sqrt{8 \cdot 1.381 \times 10^{-23} \frac{\text{J}}{\text{K}} \cdot 300 \text{ K} \cdot 1000 \Omega \cdot \left(\frac{(1 - 3 \times 10^{-4} \text{ K}^{-1} \cdot (T - T_0)) \cdot (1 + 5 \times 10^{-4} \cdot (T - T_0))}{2 + 2 \times 10^{-4} \text{ K}^{-1} \cdot (T - T_0)} \right) \cdot 1000 \text{ Hz}}}} \right)$$

The SNR of the given range is plotted below:



Task 3 c): Additional Task

What measures can be taken so that a measurement with sufficient SNR is possible in the temperature range of interest? What are advantages and disadvantages?

Answer:

Since for a balanced Wheatstone bridge the SNR is strongly dependent on the change in temperature, the wheatstone bridge can be designed for a reference temperature outside the operation range, e.g. $-100\text{ }^{\circ}\text{C}$. This will give a high and more stable SNR but introduce an offset (the bridge is unbalance in operation range).

Additionally, the SNR is a function of the bias voltage, which can be increase to increase the SNR for $T \neq T_0$, but this will increase power consumption of the bridge. Additionally, integrated circuits usually have limited supply voltages ranges.

Another way to reduce the SNR is reducing the resistance of bridge, since the signal level is independent of the resistance value but the rms-noise level is proportional to $\sqrt{R}$. If the biasing voltage is fixed, the power consumption again increases.

For most circuits, noise can be reduced by limiting the bandwidth of the system.

Part D: IS-MOS Capacitance

Task 4 a):

Fill in the gaps for characteristic of a MOS-Capacitance given in the figure below.

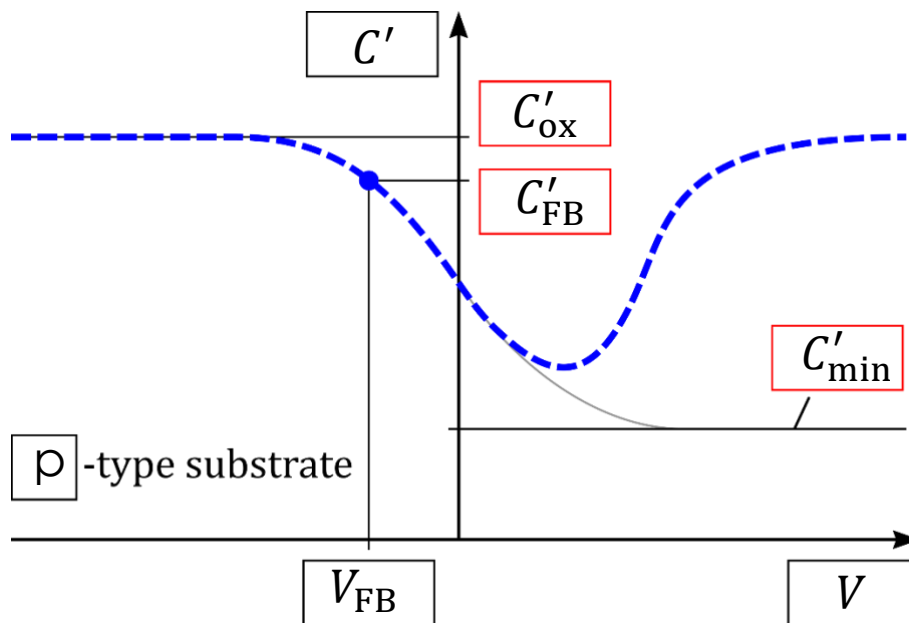


Figure 4: MOS-Cap characteristic

- i. Calculate the three characteristic parameters (red) of the curve at 300 K. Assume the following parameters:

Gate: silicon dioxide SiO_2 , $\epsilon_{\text{SiO}_2, \text{rel}} = 3.9$, $t_{\text{ox}} = 30 \text{ nm}$

Substrate: silicon Si , $\epsilon_{\text{Si}, \text{rel}} = 11.7$, $N_{\text{dopant}} = 3 \times 10^{16} \text{ cm}^{-3}$

The vacuum permittivity is $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$, the intrinsic carrier concentration in Si is $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, the elementary charge is $e = 1.602 \times 10^{-19} \text{ C}$ and the Boltzmann constant is $k_B = 1.381 \times 10^{-23} \text{ J/K}$.

- ii. Calculate how these parameters change if instead of SiO_2 the gate oxide is changed to silicon nitride ($\epsilon_{\text{SiN}, \text{rel}} = 7.5$) and the silicon substrates' doping is decreased to $0.5 \times 10^{16} \text{ cm}^{-3}$. Does this change make for a more sensitive sensor? Assume ΔV between C'_{max} and C'_{min} remains the same for both cases.

Answer:

i.

Calculation of $C'_{\max}$, C'_{FB} and $C'_{\min}$.

The maximum capacitance is given by the gate capacitance:

$$C'_{\max} = \frac{\epsilon_{\text{SiO}_2} \cdot \epsilon_0}{t_{\text{ox}}} = \frac{3.9 \cdot 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}}}{30 \times 10^{-9} \text{ m}} = 1.15 \times 10^{-3} \text{ F/m}^2$$

The flatband capacitance can be calculated from:

$$C'_{\text{FB}} = \frac{\epsilon_{\text{SiO}_2} \cdot \epsilon_0}{t_{\text{ox}} + \frac{\epsilon_{\text{SiO}_2}}{\epsilon_{\text{Si}}} \cdot \sqrt{\frac{k_{\text{B}} \cdot T \cdot \epsilon_{\text{Si}} \cdot \epsilon_0}{e^2 \cdot N_{\text{dopant}}}}}$$

$$C'_{\text{FB}} = \frac{3.9 \cdot 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}}}{30 \times 10^{-9} \text{ m} + \frac{3.9}{11.7} \cdot \sqrt{\frac{1.381 \times 10^{-23} \frac{\text{J}}{\text{K}} \cdot 300 \text{ K} \cdot 11.7 \cdot 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}}}{(1.602 \times 10^{-19} \text{ C})^2 \cdot 3 \times 10^{22} \text{ m}^{-3}}}}$$

$$C'_{\text{FB}} = 0.91 \times 10^{-3} \text{ F/m}^2$$

The minimum capacitance is given by

$$C'_{\min} = \frac{\epsilon_{\text{SiO}_2} \cdot \epsilon_0}{t_{\text{ox}} + \frac{\epsilon_{\text{SiO}_2}}{\epsilon_{\text{Si}}} \cdot \sqrt{\frac{4 \cdot \epsilon_{\text{SiO}_2} \cdot \epsilon_0 \cdot \phi_{\text{f}}}{e \cdot N_{\text{dopant}}}}}$$

With the potential

$$\phi_{\text{f}} = \frac{k_{\text{B}} \cdot T}{e} \cdot \ln\left(\frac{N_{\text{dopant}}}{n_i}\right) = \frac{1.381 \times 10^{-23} \frac{\text{J}}{\text{K}} \cdot 300 \text{ K}}{1.602 \times 10^{-19} \text{ C}} \cdot \ln\left(\frac{3 \times 10^{16} \text{ cm}^{-3}}{1.5 \times 10^{10} \text{ cm}^{-3}}\right) = 0.375 \text{ V}$$

$$C'_{\min} = \frac{3.9 \cdot 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}}}{30 \times 10^{-9} \text{ m} + \frac{3.9}{11.7} \cdot \sqrt{\frac{4 \cdot 3.9 \cdot 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}} \cdot 0.375 \text{ V}}{1.602 \times 10^{-19} \text{ C} \cdot 3 \times 10^{22} \text{ m}^{-3}}}}$$

$$C'_{\min} = 0.53 \times 10^{-3} \text{ F/m}^2$$

ii.

with the changes in material and doping the capacitances become:

$$C'_{\max} = 2.21 \times 10^{-3} \text{ F/m}^2$$

$$C'_{\text{FB}} = 0.99 \times 10^{-3} \text{ F/m}^2$$

$$C'_{\min} = 0.27 \times 10^{-3} \text{ F/m}^2$$

Assuming the voltage difference between the measurement points of $C'_{\max}$ and $C'_{\min}$ remains the same, the slope of the C-V curve in the depletion regime will

increase. This ultimately results in a more sensitive sensor, since the shift in measure capacitance per shift in voltage increases in the same way.

Task 4 b):

In the image below the blue plot gives an ideal C-V curve for a MOS-Cap with zero charges in the oxide and interface.

Sketch the low-frequency C-V-curve modifications in the plot below for:

- i. fixed (positive) oxide and interface charges (Q_{ox} and Q_{ss})
- ii. semiconductor with a higher work function
- iii. increased pH in a EOS-Cap pH sensor

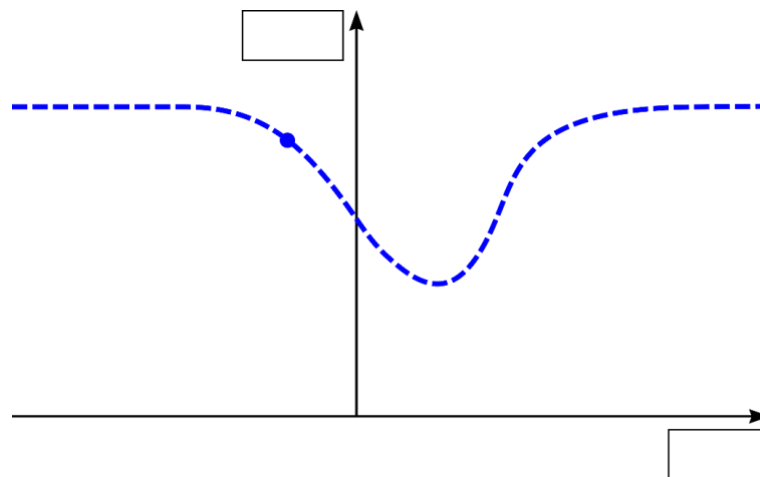


Figure 5: ideal MOS-Cap C-V curve

Answer:

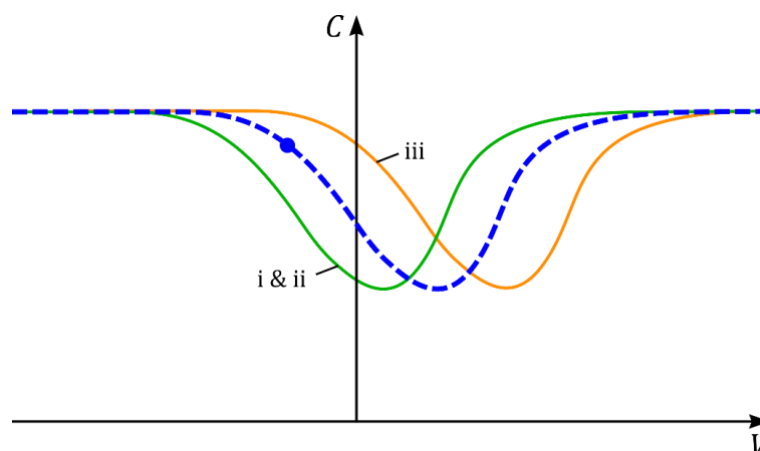


Figure 6: C-V Curve Modifications

Task 4 c):

Consider the two EOS-Cap pH-sensors with different oxide material. Calculate the sensor sensitivity ($\Delta C/\Delta \text{pH}$) assuming the devices work in a linear regime ($\Delta C/\Delta V = \text{const.}$) with identical slope for both devices. Device A has a SiO_2 gate with a sensitivity parameter of $\alpha_{\text{SiO}_2} = 0.5$ and device B has an Al_2O_3 gate with $\alpha_{\text{Al}_2\text{O}_3} = 0.9$. They work in constant voltage operation.

Answer:

For both devices the relation between measurable capacitance and voltage is given by

$$C = A \cdot V + C_0$$

The extrapolated curve will become zero at

$$V_0 = -\frac{C_0}{A}$$

and a change in pH will result in a lateral shift of the curve, equivalent to a shift in V_0 , so that

$$V'_0 = V_0 + \Delta V$$

and

$$C'_0 = -V'_0 \cdot A = -(V_0 + \Delta V) \cdot A$$

$$C'_0 = (V_0 + \Delta V) \cdot A = C_0 - \Delta V \cdot A$$

For fixed voltage operation, the measured capacitance at V becomes:

$$C' = A \cdot V + C'_0 = A \cdot V + C_0 - \Delta V \cdot A$$

$$C' = (V - \Delta V) \cdot A + C_0$$

This shows that:

$$\Delta C = C' - C = \Delta V \cdot A$$

A shift in voltage due to pH directly relates to shift in capacitance. The voltage change for an EOS-Cap can be expressed as:

$$\Delta V = \Delta \phi = -2.302 \cdot \frac{k \cdot T}{q} \cdot \alpha \cdot \Delta \text{pH}$$

The only variable parameter in this relation is α , so that:

$$\Delta C = A \cdot 2.302 \cdot \frac{k \cdot T}{q} \cdot \alpha \cdot (\text{pH}_{\text{pzc}} - \Delta \text{pH}) = -B \cdot \alpha \cdot \Delta \text{pH}$$

With $B = A \cdot 2.302 \cdot \frac{k \cdot T}{q}$, which in this case is assumed to be identical for both sensors.

The sensitivity is given by

$$S = \frac{\Delta C}{\Delta \text{pH}} = -B \cdot \alpha$$

For the devices with SiO_2 and Al_2O_3 as gate material the sensitivity becomes

$$S_{\text{SiO}_2} = -B \cdot 0.5$$

$$S_{\text{Al}_2\text{O}_3} = -B \cdot 0.9$$

Task 4 d):

How does the sensor response change if you pick another semiconductor material?

Answer:

The curve is laterally shifted due to the change in semiconductor work function. Additionally, the minimum capacitance and flatband condition are dependent on the semiconductor material, so that the slope of C-V curve in the depletion regime will eventually change as well, altering the sensor sensitivity.

Part E: Introduction to FEM Question from Unit 5

Unit 5 Example Question – *this is similar to how we give oral exam questions.*

1. Describe the 3 key steps to convert a real physical phenomena to a numerical discretization.

Answer:

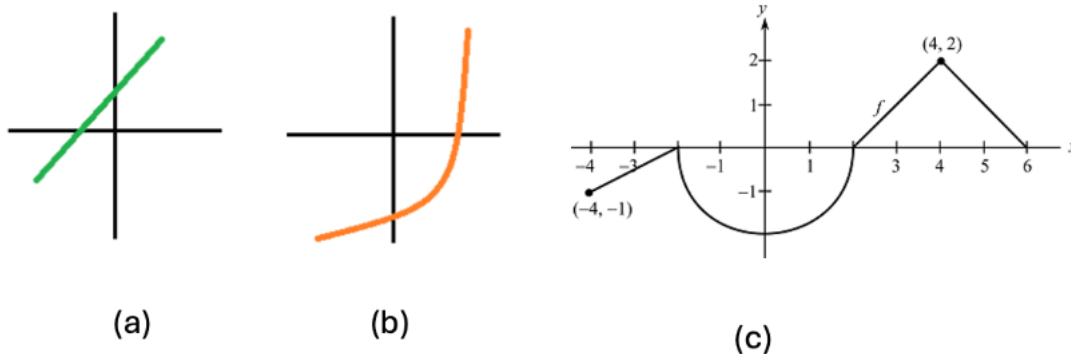
Start with a mathematical model, then convert to discretization with basis functions, that can then be solved for a numerical solution

2. What polynomial order of basis function is ideal for FEM, and explain its tradeoff with other functions with a higher or lower order.

Answer:

This depends on computational complexity and the nature of the problem. Linear basis functions (hat functions) are simpler to compute, but the tradeoff would be that more elements are needed in higher density to describe the phenomena. Higher order functions could reduce complexity of the elements with the tradeoff that it will only work in *certain physical phenomena* that suit that function type.

3. Propose a qualitative grid spacing to approximate the following curves with linear basis functions (aka hat functions).



Answer:

- a. Uniform spacing across, with a minimum of 2 points, since its linear.
- b. Wider spacing on left, smaller spacing on right during/after the bend (steeper areas need more resolution of the mesh)
- c. Uniform (or just 2 points) in linear areas, smaller spacing at sharp corners and each side of semicircle – farther spacing in the middle of the semicircle since the slope is shallow.